Selecting Reliable LLM Reasoning

DASE7506 Mini-project 2 | V6 frozen experimental report | 31 October 2026

1. Question, result and limits

Can a compact selector benefit from comparing candidate solutions, and can targeted verification improve decisions when only one candidate is correct? We combine fixed candidate representations, an out-of-fold ranking prior, conservative arithmetic evidence, question/operation alignment and two small comparison extensions. The investigation separates selecting a correct final number from verifying every reasoning step.

Official V6 test accuracy: 388/467 = 83.08351177730192%. This improves on the historical V3.1 result by three questions, but does not reach the requested score of at least 84.58%. The minimum qualifying count is **395/467 = 84.58244111349036%**, so seven more correct choices are needed. No ranking or target achievement is claimed.

Frozen method	Test correct	Accuracy
V1: two comparison networks	376 / 467	80.5139%
V2: six rich comparison networks	383 / 467	82.0128%
V3 / V3.1: rank and arithmetic	385 / 467	82.4411%
V4: question/operation alignment	383 / 467	82.0128%
V5: singleton pairwise extension	384 / 467	82.2270%
V6: low-rank semantic ranker	388 / 467	83.0835%

Data and protocol

The supplied bank contains 4,755 training, 798 validation and 467 test questions, each with four candidate texts and fixed 1,024-dimensional representations. Numeric equivalence is determined by the unchanged official parser. Multiple correct candidates can earn the same point. There are 2,114 hard training and 355 hard validation questions, defined by at least two distinct valid parsed answers. All test groups are hard according to the course guide; easier validation groups must not inflate the main comparison.

Weights, vocabularies, input normalisation and projections are fitted using supplied training data only. Validation selects architectures, epochs, mixing coefficients and score-dispersion scales. Each reported version is frozen before scoring. Earlier public test aggregates were known during later development; test references and per-example test errors were not inspected for development. This is an exposed public benchmark, not an untouched holdout. An independent audit remains necessary.

The frozen comparison prior

2. Candidate representations and training

The basic 24 features describe parse validity, same-answer count, length, formatting, numeric counts, signed log answer magnitude, sign/zero/integrality, distinct-answer count and ten discourse-word counts. Rich features expand this to 206 dimensions with question quantities, question/solution word counts, punctuation, answer-relative values and conservative equation checks. No question ID or candidate-position embedding is supplied to a learner.

Three rich networks read [embedding, features], dimension 1,230; three delta networks read [embedding, embedding minus group mean, features, features minus group mean], dimension 2,460. A shared Linear(d,192)-LayerNorm-GELU-Dropout(0.3) encoder produces h. With group mean m, the head applies Linear(576,96)-GELU-Dropout(0.25)-Linear(96,1) to [h,h-m,m], plus a 0.1 linear residual. Training standardisation uses floors 0.1 for rich and 0.03 for delta inputs, then clips to [-15,15].

Candidate BCE, pairwise ranking and correct-set mass objectives were compared. The selected mass loss is logsumexp(all scores) minus logsumexp(correct scores), plus 0.25 BCE. It rewards any correct answer, matching the scorer. Rich seeds 300-302 use AdamW 0.0004, decay 0.025 and 65 epochs; delta seeds 400-402 use 0.0003, decay 0.04 and 55 epochs. Batches contain 128 questions. Hard groups have weight 3; easy groups have weight 1 for rich and 0.25 for delta.

Selected family / seeds	Epochs (zero-based)	Weight
Rich mass: 300, 301, 302	19, 3, 6	0.25
Delta mass: 400, 401, 402	52, 19, 28	0.75

Each family is divided by a saved hard-validation within-question dispersion, 2.9112713 and 6.8261561 respectively. Answer pooling mixes half of a candidate score with half of the mean score for candidates sharing its parsed answer. Invalid final answers receive a masking sentinel. These scores are ranking evidence, not calibrated probabilities.

Out-of-fold ranking and arithmetic

Four shuffled training folds (seed 1000) create held-out predictions from one rich and one delta mass network per fold, each trained for 24 epochs with fold-only statistics. A 422-feature ranker input concatenates the normalised prior, nine arithmetic features, 206 rich features and their 206 centred differences. Two YetiRank models fit hard training groups: seeds 1700/1701, depths 4/5, learning rate 0.025, L2 20 and 32 borders. Validation selects 200/175 trees from runs of 1,600, mean rank multiplier 2 and arithmetic coefficient 0.5.

A restricted syntax-tree interpreter checks numeric equations and propagates conservative repairs; unsupported syntax is skipped and candidate code is never executed. Training repair proposals achieved 79.75% accuracy on 163 triggered hard groups, insufficient for unconditional overrides. The bonus is 0.5 times (repair support minus 0.5 times contradiction count). Meta features are rounded to four decimals for numerical stability. Arithmetic consistency cannot establish that the right word problem was solved.

Targeted verification and semantic ranking

3. Question alignment and singleton comparison

V4 adds a question TF-IDF encoder: 3,000 training-fitted unigram/bigram features, with numeric mentions replaced by positional tokens, project to 64 dimensions. Candidate embeddings, 24 basic features and a 32-dimensional reconstructed operation profile project to 128 dimensions. The scoring head compares candidate, group, question and operation interactions. Restricted arithmetic parsing produces full or partial candidate profiles; reference profiles supervise only the training auxiliary task.

Six runs compare auxiliary supervision on/off at seeds 2500-2502. AdamW uses learning rate 0.0005, decay 0.03, batches 128, 65 epochs, hard/easy weights 3/0.5 and mass loss plus 0.25 BCE; auxiliary MSE has weight 0.2. Validation selects auxiliary seed 2500, epoch 6, mixing 0.25 and dispersion 2.3710036. V4 gains two validation questions but loses two test questions, demonstrating that additional supervision does not ensure transfer.

V5 uses a 64-dimensional shared pairwise network. Inputs are the embedding (1,024), its difference from the valid-candidate mean (1,024) and 24 basic features; answer-count and distinct-answer features are zeroed. Input statistics use hard valid training candidates, with standard-deviation floors 0.1/0.02/0.1 and clipping to [-10,10]. For encodings h_i,h_j, a 256-to-64-to-1 head reads [h_i,h_j,h_i-h_j,h_i*h_j]. Antisymmetrisation yields r_ij = -r_ji, including a small linear residual. A candidate score averages comparisons against valid candidates.

The selected pairwise model trains on the **684 training questions with exactly one correct candidate**. It minimises mean softplus(1-r_ij) for correct/incorrect valid pairs. Seed 3906, hidden size 64, AdamW 0.0004, decay 0.1, batches 96 and seven epochs are selected from eight 60-epoch direct-pair runs. Its saved scale is 1.3169010 and mixing is 0.25. Validation singleton accuracy rises from 83/114 to 85/114, while two other hard choices are harmed: overall hard accuracy remains 320/355.

The V6 low-rank semantic ranker

A training-fitted 64-component PCA projects within-question embedding differences. The 510 features comprise two prior/confidence features, 32 operation features, 206 rich features, 206 centred differences and 64 semantic components. Training uses a four-fold (seed 3200) tree correction on existing neural out-of-fold scores; inference prior features use V5. This is an approximate stacked feature pipeline, not a fully nested, cross-fitted V5 system. The train/inference prior mismatch and indirect dependence across stacking folds are limitations.

Four GPU runs compare depths 4/5, prior initialisation on/off and singleton weights 1/3. Validation selects seed 4103, depth 5, PairLogit, learning rate 0.025, L2 30, 32 borders, singleton group weight 3, no initial baseline logit, and 125 trees from an 800-tree trajectory. The input still contains the out-of-fold prior. Scores are divided by 1.0142421 and added to V5 with weight 0.25 after half same-answer pooling. This final branch improves validation by one question and test by four. It does not isolate PCA as the sole cause of the gain.

Experiments, ablations and rejected ideas

4. Comparisons under the same official protocol

Validation method	All / 798	Hard / 355
Parsed-answer majority vote	86.22%	69.01%
Basic-feature logistic baseline	90.10%	77.75%
Embedding logistic baseline	89.35%	76.62%
V2 six-network mass prior	94.49%	87.61%
V3.1 rank + arithmetic prior	761 (95.36%)	318 (89.58%)
V4 question alignment	763 (95.61%)	320 (90.14%)
V5 direct-pair extension	763 (95.61%)	320 (90.14%)
V6 semantic ranker added	764 (95.74%)	321 (90.42%)

Removing only the final semantic branch returns the frozen V5 system. This is the key component ablation: V5 to V6 adds one hard-validation success (85 to 86 singleton successes) and four test successes. The smaller public test differences are descriptive; no significance claim follows from them. Different families below have separate searches and are not matched causal ablations of an information source.

Additional experiment family	Scope	Development outcome
Program / profile predictors and rankers	8 nets + 6 trees	No useful fusion
Sequence / dynamic transformers	3 + 5 runs	Not selected
CV quality nets / MLM fine-tuning	16 configs; 3 runs	No useful fusion
Relational / direct / rich-pair networks	8 + 8 + 4 runs	Direct pair selected
PCA ensembles / PLS rankers	6 + 4 runs	V6 still preferred
Geometry / coverage / focused trees	4 + 4 runs	No further gain

The operation probe reached 70.26% training out-of-fold final-operation accuracy versus a 33.54% majority baseline, yet did not improve answer selection. A CPU cross-encoder trial was stopped and is not counted as a completed comparison. Candidate program coverage, failed formula transfer, semantic-rule precision and every search setting are retained in code/v4/records. No rejected checkpoint is required for inference.

The final robustness criterion was defined before the V5 test: maximise singleton-validation accuracy while retaining at least 319/355 hard successes; ties favour hard/all accuracy and smaller models. Later richer averages and projections did not beat V6 under that criterion and were not tested. Extensive reuse of 355 hard and 114 singleton validation examples can still select noise.

Reliability beyond an accuracy number

5. Answer-group confidence

Top-two candidate margins are misleading when candidates share an answer or invalid answers are masked. We instead take the maximum score within each distinct valid numeric-answer group, then subtract the runner-up group score. Only the 355 hard validation questions enter the following quartiles, sorted by increasing margin. No reference answer is needed to compute the margin.

V6 answer-group margin quartile	Correct / count	Accuracy; Wilson 95%
Lowest: 0.009 to 2.031	73 / 89	82.02%; 72.77-88.62
Second: 2.037 to 3.453	76 / 89	85.39%; 76.60-91.26
Third: 3.454 to 4.398	84 / 89	94.38%; 87.51-97.58
Highest: 4.399 to 6.451	88 / 88	100%; 95.82-100

The upper half reaches 172/177 = 97.18% accuracy at 49.86% coverage of hard validation. This supports using group margins as a descriptive confidence ranking. A margin of 4 is not a probability, and 88/88 does not establish perfect reliability. Validation reuse makes these estimates optimistic; calibration and deployment thresholds require an independent set. The submission still chooses an answer for every question; abstention is only an analysis device.

Small changes and uncertainty

Relative to V5, V6 corrects two hard-validation choices and harms one: net +1/355, or +0.28 percentage points. A paired question bootstrap with 10,000 draws and seed 4300 gives a conditional 95% interval of -0.56 to +1.13 points. On singleton questions it corrects one and harms none: +0.88 points, with bootstrap interval 0 to +2.63. These intervals condition on already selected models and do not account for the full model search.

The pairwise extension similarly corrected two singleton choices but harmed two other hard choices. This shows why improving a difficult subgroup does not imply an overall gain. Observable all-distinct-answer gating identifies 54 validation questions, all singletons, but routing only those cases through the extension did not improve the selected criterion. We therefore did not deploy that gate.

What remains unreliable

V6 makes 34 validation errors and 79 test errors. Hard-validation accuracy is 90.42%, versus 83.08% test accuracy: a 7.34-point gap. The gap is consistent with development selection optimism and difficulty differences, but these experiments cannot separate their causes. No claim about test label multiplicities is based on inspecting test references.

Arithmetic can flag a bad sum or fraction conversion; internally consistent solutions can still omit a condition, confuse revenue with profit, or return an intermediate quantity. A correct number may also coexist with an invalid derivation. Fixed representations and sparse training supervision remain imperfect evidence of semantic correctness. More complex auxiliary models mostly failed to improve this limitation.

Resource use and reproducibility checks

6. Complete CPU inference budget

Measured V6 quantity	Value	Budget / context
All uncompressed inference assets	13,031,770 bytes	12.428 MiB < 16
Fresh prediction + separate scoring	9.694 seconds	Local < 30 seconds
Runner prediction time	8.768 seconds	Startup excluded
Official scorer time	0.030 seconds	Separate process
Inference parallelism	Four threads	All libraries limited

Local hardware is Intel Core i7-12700H, Windows 11. Inference uses Python 3.13.5, NumPy 2.2.6, PyTorch 2.7.1+cpu, scikit-learn 1.6.1 and CatBoost 1.2.10. The measured interval includes fresh process startup, imports, asset/data loading, all preprocessing and every ensemble member. The reference Xeon Platinum 8457C and course-recommended Python 3.12 were not available for verification; local timing alone does not prove reference-machine eligibility.

The final directory contains eight neural models and three tree models, their input transformations, one vocabulary and the configuration. The asset total counts every file in runs/final, including unused configuration metadata. No model weights are downloaded at prediction time. Source files and supplied data are excluded from this permitted-asset count. Training/development caches are separately labelled and are not read by inference.

Training and search costs

Recorded family wall time	Seconds	Training device
Rich / delta network searches	310.5 / 324.8	CPU
Alignment six-run search	152.3	CPU
Direct-pair eight-run search	~60	RTX 3060 Laptop
Semantic ranker four-run search	101.1	RTX 3060 Laptop
PCA ensemble / MLM fine-tuning	155.6 / 317.1	RTX 3060 Laptop
Selected V5 / V6 CPU retraining	5.0 / 63.5	CPU

The laptop GPU has 6 GB memory and is below the allowed RTX 4090 limit. It was used only for training; no additional pretrained weights were obtained. Neural training from scratch and in-domain language pretraining use only supplied training data. CUDA runtime installation is a software dependency, not a pretrained language model. Family times include different preprocessing/search scopes and some jobs overlap, so they must not be summed as a total project wall time.

Official data hashes and evaluator files were checked. The released checkpoint reproduces the recorded prediction choices. Eleven validation questions with new IDs and candidate permutations preserve selected numeric answers. Four-decimal tree input rounding limits sensitivity to small floating-point changes; this finite smoke check is not a proof for every input. Separate CPU retraining smoke runs completed for both selected extensions; cross-device retraining need not reproduce the frozen weights or score exactly.

Reproduce, disclose and submit

7. Frozen result reproduction

From code/, install requirements.txt in an isolated Python environment, then set OMP_NUM_THREADS, OPENBLAS_NUM_THREADS, MKL_NUM_THREADS and NUMEXPR_NUM_THREADS to 4. The main README supplies exact PowerShell commands. No retraining is needed to reproduce the claimed score.

```
python check_data.py

python predict.py --method submission_v6.py --checkpoint runs/final --split test --output
runs/reproduced-v6-predictions.json

python score_predictions.py --split test --predictions runs/reproduced-v6-predictions.json --output
runs/reproduced-v6-score.json
```

Expected official accuracy is **83.08351177730192**. runs/v6-test-score.json is the official result, v6-resources.json records complete external timing, v6-checks.json records checks, and frozen_v6.json identifies all frozen inference assets and source hashes. Its test_evaluated=false field describes the state at freeze time; subsequent official scoring is recorded separately. It is not a claim that the reported version has never been tested.

Training reproduction

train_all.ps1 replays original baseline development using the supplied training/validation splits. train_v4.ps1 rebuilds the alignment extension in a separate output. train_v5.py --device cpu --output runs/retrained_v5 rebuilds the selected pairwise extension; --device cuda uses the GPU training path. train_v6.py --device CPU --output runs/retrained_v6 rebuilds the semantic ranker while retaining the supplied frozen baseline; --device GPU uses its GPU path. Both selected-extension CPU commands were executed successfully locally.

The training-only neural out-of-fold cache is retained under runs/v2 and can be regenerated by train_stack.py. build_v3_oof.py recreates the next-stage training cache. code/v4 contains research scripts and detailed records for positive and negative studies. Development scripts may require preceding feature-building steps; TRAINING.md lists their order. Fixed released weights, rather than cross-device retraining, are the claimed-score reproduction target.

AI, reuse and submission status

OpenAI Codex substantially assisted with architecture and feature design, code, training, experiment selection, analysis and report preparation. The student must review and understand the implementation and results. Supplied evaluation tools and data were retained unchanged. GSM8K-derived questions and references retain the supplied MIT license and attribution. Candidate solutions and fixed representations were prepared by the course using Qwen3-0.6B. No external datasets or additional pretrained weights were used.

No public repository, checkpoint URL or GitHub submission issue has been created. Student ID and a public destination have not been supplied. Final submission requires code and checkpoint/configuration links, then actual creation of the GitHub issue prepared by the course website before **31 October 2026, 23:59 UTC+8**. If submitting this V6 version, self-report its official test score above, with submission_v6.py and the matching runs/final checkpoint. Validation accuracy must not be self-reported as test accuracy.

Sources and evidence

Course guide | Official running instructions | Submission website. Numerical evidence is in the bundled official scores, freeze manifests, reliability analysis and experiment records. This report contains seven pages.